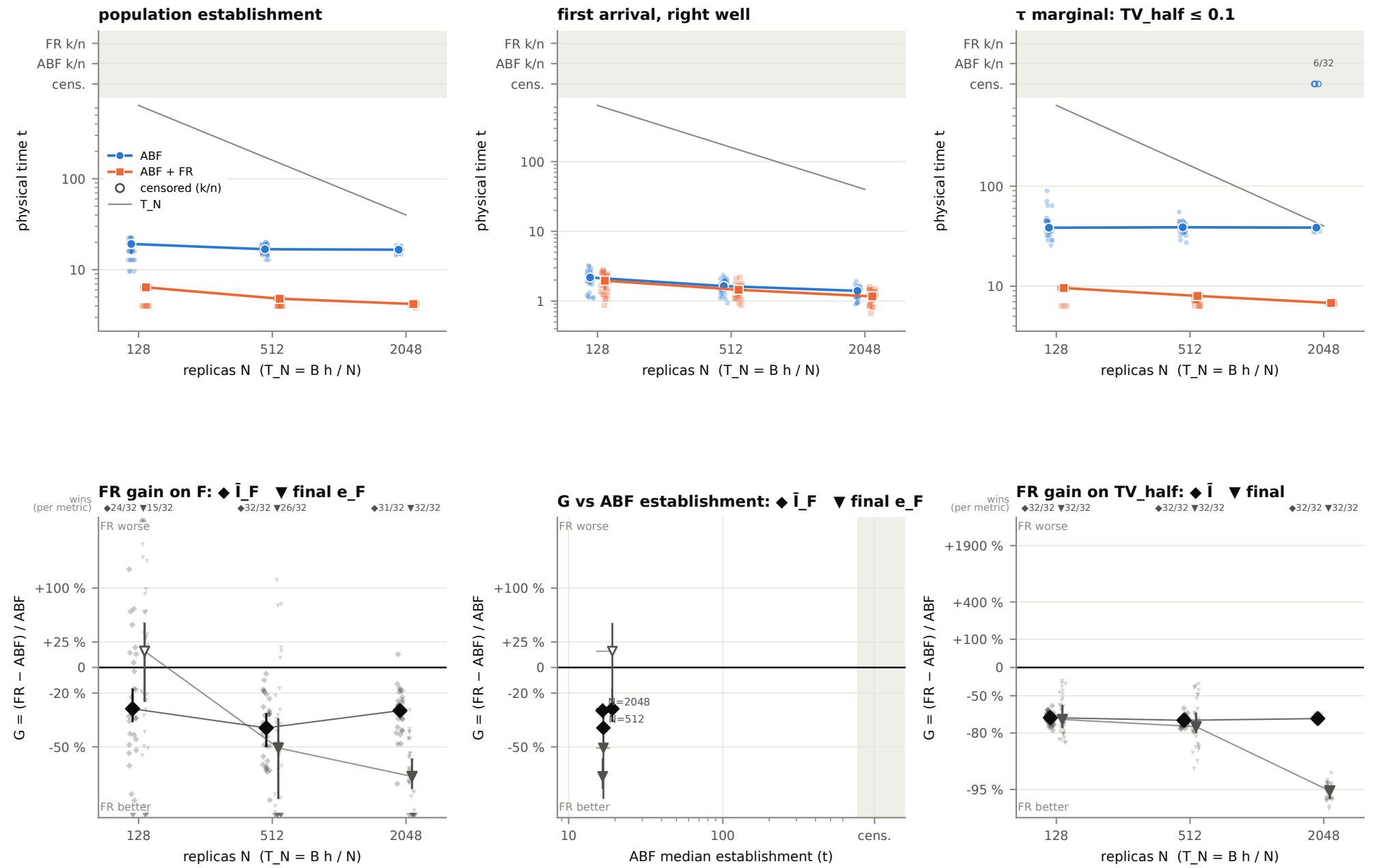


gateway alpha1: population establishment, first arrival and marginal convergence vs N (physical time)

Equal total budget per arm per seed: $B = N \times n_steps = 3.2768e+09$ walker-steps at every N ($h = 2.5e-05$, $T_N = B \cdot h / N$); 32 seeds per (N , method). Do FR gains coincide with establishment-limited N ?



FR gain on F: ♦ $\bar{I}_F$ ▼ final e_F

wins (per metric) ♦24/32 ▼15/32 ♦32/32 ▼26/32 ♦31/32 ▼32/32

G vs ABF establishment: ♦ $\bar{I}_F$ ▼ final e_F

FR worse

FR better

10 100 cens.

FR gain on TV_half: ♦ $\bar{I}$ ▼ final

wins (per metric) ♦32/32 ▼32/32 ♦32/32 ▼32/32 ♦32/32 ▼32/32

gateway alpha1 · S6b_establishment_time · status: all planned (N, seed, method) complete · FR is not defined at $N = 1$ (ABF only, by design).
Establishment = first save after which the instantaneous far-state fraction (right well $x > 0.5$) stays $\geq \frac{1}{2} \times 0.361$. Shaded N []: the instantaneous rule is UNRELIABLE there (ill-defined at $N \leq 2$; an exactly uniform population passes every save of the second half with probability < 0.9 , summary establishment_null), so the same rule on the cumulative visitation is plotted; FR birth-death can push the instantaneous count below binomial scatter, so an FR establishment advantage at those N is not evidence of faster convergence. $\tau(TV_half)$ on the budget axis uses the common budget grid.
Top: seed median (large open marker in the 'cens.' band = median censored), IQR bars, faint seeds. Bottom: paired G with bootstrap 95 % CIs (filled = excludes 0; 'wins' = $FR < ABF / n$); scatters: each N 's G against the ABF median time (horizontal bar = ABF IQR; 'cens.' = censored).
Source results/mechanism/matched_free_energy/alpha1/analysis/summary.json (generated 2026-10-10T14:19:18Z, eqb_metrics/2, 10000 bootstrap resamples); plot_synthesis/2.